Brain-Shapelet: A Framework for Capturing Instantaneous Abnormalities in Brain Activity for Autism Spectrum Disorder Diagnosis

Yijie Ren, Zhengwang Xia, Yudong Zhang, *Senior Member, IEEE*, and Zhuqing Jiao

***Abstract*— Some symptoms of Autism Spectrum Disorder (ASD), such as anxiety and depression, often manifest intermittently rather than continuously, complicating the identification of reliable pathophysiological biomarkers. Meanwhile, functional connectivity networks (FCNs) generate high-dimensional connectomes, making it difficult to accurately capture instantaneous abnormal biomarkers of neurological disorders. To address this issue, we propose a framework, called Brain-Shapelet, to extract discriminative subsequences (Shapelets) from functional magnetic resonance imaging (fMRI) data for capturing instantaneous abnormalities in brain activity. It applies random walk algorithm on group-representative brain network to obtain brain region sets, and aggregates their blood oxygen level-dependent (BOLD) signals to extract Shapelets that reflect the associations between different brain regions at the same time point. Specially, we develop a feature selection strategy to reduce redundancy in Shapelets and optimize classification performance. Brain-Shapelet places greater emphasis on short-term brain activity alterations, allowing it to capture instantaneous abnormalities more effectively. It is evaluated on the ABIDE dataset and achieves a classification accuracy of 82.8%, significantly outperforming traditional brain network modeling methods. The proposed co-occurrence rate, occurrence frequency, and Gini coefficient metrics quantify the contributions of brain regions from the perspective of Shapelets, offering valuable insights for ASD diagnosis.**

***Index Terms*— Brain-Shapelet, autism spectrum disorder, functional magnetic resonance imaging, instantaneous abnormalities, Shapelet extraction.**

I. Introduction

AUTISM Spectrum Disorder (ASD) is a complex neurodevelopmental disorder characterized by pronounced variations in social communication and behavioral patterns among individuals [1]. Globally, ASD affects approximately 1 in 36 children, presenting substantial long-term health challenges for children and their families [2]. Early diagnosis and intervention have markedly improved outcomes for individuals with ASD, with earlier detection resulting in even greater benefits [3]. However, the current diagnostic approaches for ASD lack clear pathophysiological biomarkers, relying primarily on psychological criteria [4]. This behavior-based and subjective diagnostic method is prone to the influence of individual clinicians' opinions. Consequently, there is an urgent need to develop objective tools that can assist clinicians in achieving more accurate and consistent ASD diagnoses.

Functional magnetic resonance imaging (fMRI) is a non-invasive technique that captures changes in blood oxygen level-dependent (BOLD) signals driven by neuronal activity [5]. It enables researchers to delve into integration pattern differences between ASD and normal control (NC) groups, providing robust support for identifying the potential causes of neurological disorders [6]. Nowadays, fMRI has been extensively applied to uncover biomarkers for a variety of neurological conditions, such as frontotemporal dementia, Alzheimer diseases, and major depressive disorders [7].

Currently, it remains a key research focus to construct brain network models in neurological disorders diagnosis. The mainstream methods of brain network modeling involve the linear associations of BOLD signals across different brain regions to construct functional connectivity networks (FCNs) [8]. These strategies can be broadly classified into two categories: static brain networks and dynamic brain networks [9], [10]. The former assumes that functional connectivity(FC) strength between brain regions remains constant during the sampling period. Numerous studies have proved that this assumption does not align with reality [11]. In contrast, the latter depends on a fixed sliding window with specific step sizes, generating

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a series of time-varying FCNs, collectively referred to as connectomes [12]. Importantly, clinical studies have indicated that many symptoms of neurological disorders, such as anxiety and depression, often exhibit intermittent characteristics rather than persistent manifestations [13]. It suggests that biomarkers of neurological disorders are more likely to be instantaneous abnormal changes rather than stable or continuous patterns of abnormal connectivity [14]. In summary, it is extremely difficult to extract instantaneous biomarkers associated with neurological disorders from high-dimensional connectomes.

Differing from current mainstream brain network modeling methods, Shapelets are the discriminative subsequences of time-series that are the most predictive of the target variable [15]. They have been proven effective on capturing discriminative patterns between different categories of time-series, providing explanations for instantaneous anomalies in time-series data [16], [17], [18], [19], [20]. For instance, Dai et al [21] proposed a Shapelet-based EEG channel selection method to optimize brain-computer interface (BCI) applications. Chen et al [22] extracted HF-HRV segments from electrocardiograms (ECG) through Shapelet analysis to capture patterns of the autonomic nervous system activity related to learning. He et al [23] developed ShapeWordNet, an interpretable Shapelet neural network that transforms complex physiological signals into discrete, meaningful word sequences. Similarly, Janyalikit and Ratanamahatana [24] adopted Shapelets to extract representative subsequences from EEG signals to identify spontaneous ankle dorsiflexion movement intentions. The practicality and interpretability of Shapelets have been well-validated across various studies [25], [26].

Nevertheless, there are several challenges in extracting Shapelets from fMRI data. The BOLD signal of each brain region can be regarded as a univariate time-series. Existing studies mainly focus on extracting Shapelets from such univariate data, leading to inconsistencies in the temporal scales of Shapelets. These inconsistencies cause the loss of association s between different brain regions at the same time point, limiting the applicability of Shapelets to the multidimensional nature of fMRI.

For the aforementioned issues, we proposes Brain-Shapelet, a framework for extracting Shapelets to capture instantaneous abnormalities in brain activity for ASD diagnosis. It constructs group-representative brain networks with sparse FCNs and applies random walk algorithm to obtain brain region sets. Then, it aggregates the BOLD signals of brain region sets to extract Shapelets. This process aligns BOLD signals from different brain regions to the same timescale, ensuring Shapelets effectively capture the associations among the brain regions. Finally, the feature selection strategy filters the Shapelets after combining the Jaccard similarity coefficient and mutual information. In addition, we define a series of metrics—co-occurrence rates, occurrence frequencies, and Gini coefficients to measures the roles of brain regions. The main contributions of this paper are summarized as follows:

(1) Brain-Shapelet selects Shapelets according to discriminative ability, thereby reducing feature dimensionality to more effectively capture instantaneous anomalies in brain activity than high-dimensional connectomes.

(2) The interpretable analysis measures the roles of brain regions from the Shapelet perspective, contributing to the exploration of potential instantaneous biomarkers.

(3) The experimental results evaluated Brain-Shapelet on the ABIDE dataset and proved that the framework outperforms FCN-based methods in ASD diagnosis, validating its effectiveness.

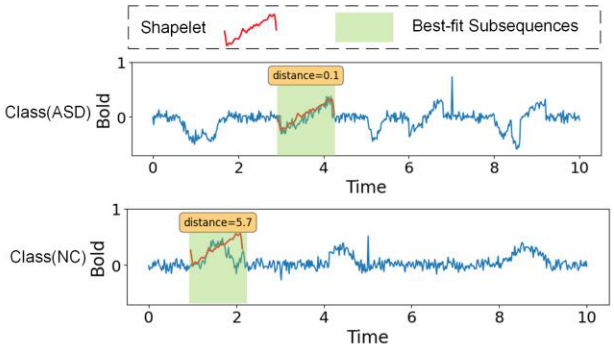


Fig. 1. Shapelet in a univariate time series. The best-fit subsequence is the subsequence within the time series that has the smallest distance to the Shapelet.

II. Guidelines for Manuscript Preparation

This section introduces the process of Brain-Shapelet extracting Shapelets from fMRI data to capture instantaneous abnormalities in brain activity (Sec. A). A feature selection strategy is designed to address the redundancy issue of Shapelets (Sec. B). the roles of brain region (Sec. C) Finally, are measured to perform interpretability analysis on the extracted Shapelets.

Fig.1 shows the concept of Shapelets in univariate time-series data. Compared to FCN-based methods, Shapelets focus on local patterns within time-series data, capturing instantaneous changes in brain activity. This focus not only helps mitigate the impact of physiological signal noise [27], but also provides robust interpretability—each Shapelet represents an inherent pattern within the time-series data, offering intuitive insights into the classification results [28].

Fig.2 shows the detailed workflow of Brain-Shapelet, which is divided into two main parts. The first part involves obtaining brain region sets by random walk algorithm on the group-representative brain network, corresponding to Steps (1)–(3). The second part involves extracting Shapelets from the brain region sets, corresponding to Steps (4)–(5).

A. Brain-Shapelet

BOLD signals across brain regions typically exhibit functional correlations. Extracting Shapelets from a single brain region may be insufficient to capture the coordinated activity patterns between brain regions. Therefore, this study jointly extracts Shapelets from the BOLD signals of functionally correlated local brain region sets to comprehensively identify discriminative patterns that distinguish ASD from NC subjects.

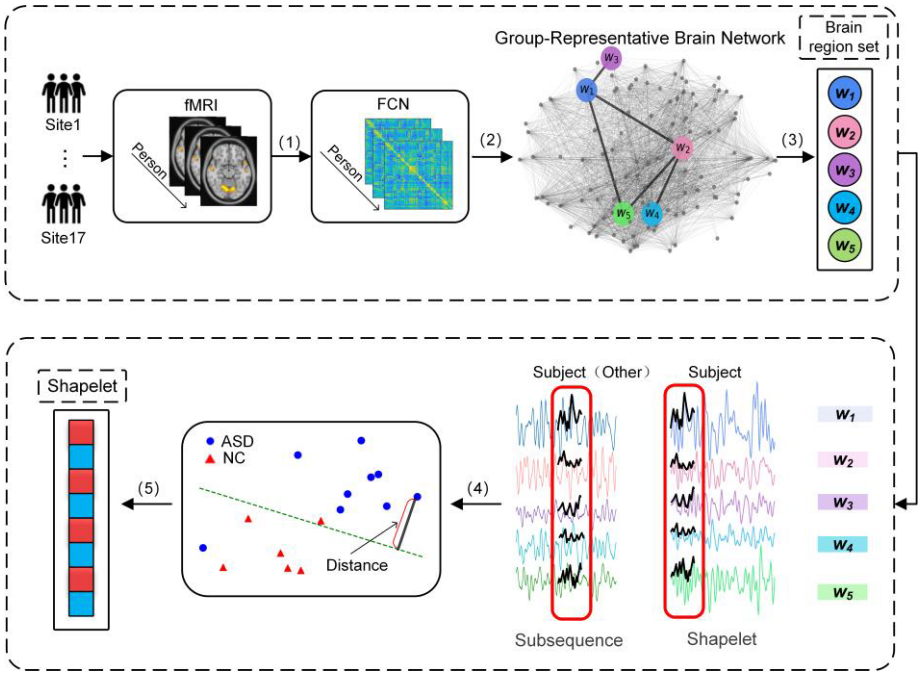


Fig. 2. The framework of Brain-Shapelet. (1) Constructing an FCN with Pearson correlation and sparsification; (2) Integrating individual FCNs into the group-representative brain network; (3) Obtaining brain region sets by random walk algorithm; (4) Identifying an optimal splitting point; (5) Extracting Shapelet based on the classification accuracy of the optimal splitting point.

The FC of different subjects shows notable variability resulting from individual differences and noise. To address this issue, a group-representative brain network is constructed to integrate the consistent connections among all subjects, improving data consistency and minimizing the interference of noise. Assume there are U subjects in total, the fMRI data of subject u can be divided into Nv brain regions on a specific template. Let $\boldsymbol{BOLD}_u = \{\boldsymbol{B}_1, \ldots, \boldsymbol{B}_{Nv}\}$, $\boldsymbol{B}^u_{Nv} = \{b^u_{Nv,1}, \ldots, b^u_{Nv,L}\}$ is the BOLD signal of the Nv brain region for subject u, and the length of each signal is L. A sparse FCN, denoted as $\boldsymbol{F}^u \in \mathbb{R}^{Nv \times Nv}$, is constructed by the BOLD signals of different brain regions, to describe the interactions between brain regions. The functional connections between brain region i and brain region j is determined by the Pearson correlation coefficient $Per(\cdot)$ as follows:

$$\boldsymbol{F}^u_{i,j} = \begin{cases} \left|Per(\boldsymbol{B}_i, \boldsymbol{B}_j)\right|, & \left|Per(\boldsymbol{B}_i, \boldsymbol{B}_j)\right| \geq \beta \\ 0, & \left|Per(\boldsymbol{B}_i, \boldsymbol{B}_j)\right| < \beta \end{cases} \tag{1}$$

where the threshold β sparsifies the FCN, removing weakly correlated functional connections to suppress noise.

The consensus matrix $\boldsymbol{CM} \in \mathbb{R}^{Nv \times Nv}$ is defined for all subjects' $\boldsymbol{F}_{i,j}$. In this matrix, each element $\boldsymbol{CM}_{i,j}$ indicates the frequency of $\boldsymbol{F}_{i,j} \neq 0$ across all subjects:

$$\boldsymbol{CM}_{i,j} = \frac{1}{U} \sum_{u=1}^{U} \delta(\boldsymbol{F}^u_{i,j} \neq 0) \tag{2}$$

where $\delta(\boldsymbol{F}^u_{i,j} \neq 0)$ is an indicator function, which takes the value 1 when $\boldsymbol{F}^u_{i,j} \neq 0$, indicating that the functional connection exists between brain regions i and j for subject u; otherwise, it takes the value 0.

The group-representative brain network $\boldsymbol{F}^{group} \in \mathbb{R}^{Nv \times Nv}$ is constructed by the elements of consensus matrix $\boldsymbol{CM}_{i,j}$.

$$\boldsymbol{F}^{\text{group}}_{i,j} = \frac{1}{|\boldsymbol{CM}_{i,j>\sigma}|} \sum_{u \in \boldsymbol{CM}_{i,j>\sigma}} \boldsymbol{F}^u_{i,j} \tag{3}$$

where the threshold σ select high-frequency functional connections. This method better reflects the consistency of functional connections across subjects [29].

In $\boldsymbol{F}^{\text{group}}_{i,j}$, A brain region sets $\boldsymbol{W} = \{w_1, \ldots, w_k\}$ is generated after k iterations of the random walk algorithm to simulate local inter-regional connectivity patterns. Each $w_{\text{i}} \in \boldsymbol{W}$ represents the index of a brain region visited during the walk. Starting from an initial region w_1,the walk proceeds to a neighboring region w_{i+1} with a transition probability as follows:

$$P(w_{i+1}|w_i) = \frac{\boldsymbol{F}^{group}_{w_i,w_{i+1}}}{\sum_{e \in \boldsymbol{N}(w_i)} \boldsymbol{F}^{group}_{w_i,e}} \tag{4}$$

where $Ner(w_i)$ represents the set of all neighboring nodes of brain region w_i.

The BOLD signal of brain region Nv is denoted as $\boldsymbol{B}_{Nv} = \{b^{Nv}_1, \ldots, b^{Nv}_L\}$. The collection of subsequences $\boldsymbol{T}$ of length l, starting at position 0, can be written as $\boldsymbol{T}_{Nv} = \{\boldsymbol{t}_{Nv,1}, \ldots, \boldsymbol{t}_{Nv,N}\}$, where $\boldsymbol{t}_{Nv,n} = \{b^{Nv}_{(n-1)l}, \ldots b^{Nv}_{nl}\}$, resulting in a total of $N : L - l + 1$ subsequences, different subjects may have different L. The subsequence at position n is extracted along the brain region set $\boldsymbol{W}$ from $\boldsymbol{T}_{Nv}$, and define a subsequence set $\boldsymbol{S}_{\boldsymbol{W},n} = \{\boldsymbol{t}_{w_1,n}, \ldots, \boldsymbol{t}_{w_k,n}\}$. Each subject has such N subsequence sets, representing local brain activity patterns. A subsequence set $\boldsymbol{S}^u_{\boldsymbol{W},i}$ at position i from subject u was selected as a Shapelet candidate, and compare it with

the subsequences set of subject r for similarity. The best-fit subsequence set in subject r is then identified by computing the shortest Euclidean distance $m_{u,r}$ to the Shapelet candidate.

$$m_{u,r} = distance(\boldsymbol{S}^u_{\boldsymbol{W},i}, \boldsymbol{S}^r_{\boldsymbol{W},n}) = \min_{n=1,\ldots,N} \sum_{k=1}^{K} (\boldsymbol{t}^u_{w_k,i} - \boldsymbol{t}^r_{w_k,n})^2 \tag{5}$$

Euclidean distances of all subjects are combined into a set $\boldsymbol{M} = \{\boldsymbol{m}_{u,1}, \ldots, \boldsymbol{m}_{u,U}\}$, The goal is to find an optimal splitting point d^*, that effectively separates ASD subjects from NC subjects within $\boldsymbol{M}$. Shapelets are selected according to the classification accuracy of d^*.

$$d^* = \arg\min_d \left(\frac{|\boldsymbol{M}^0|}{|\boldsymbol{M}|} \cdot \boldsymbol{Gain}(\boldsymbol{M}^0) + \frac{|\boldsymbol{M}^1|}{|\boldsymbol{M}|} \cdot \boldsymbol{Gain}(\boldsymbol{M}^1)\right) \tag{6}$$

$$Gain(M) = 1 - (num^2_{NC} + num^2_{ASD}) \tag{7}$$

where $\boldsymbol{M^0}$ is the subset of $\boldsymbol{M}$ when $\boldsymbol{m} < d^*$, and $\boldsymbol{M}^1$ is the subset of $\boldsymbol{M}$ when $\boldsymbol{m} \geq d^*$. num_{NC} and num_{ASD} indicate the proportions of NC and ASD subjects in their respective subsets, respectively. The Gini Impurity $Gain(\bullet)$ measures the separation quality of the d^*.

The Shapelet distance feature $\boldsymbol{x}$ is defined by comparing the number of ASD subjects num^0_{ASD} and NC subjects num^0_{ASD} in $\boldsymbol{M^0}$, aligning d^* to 0. For $\boldsymbol{x}$, values less than 0 indicate a higher likelihood of NC subjects, while values greater than 0 suggest a higher likelihood of ASD subjects. Values closer to 0 imply the presence of similar subsequences corresponding to the Shapelet in the samples. $\boldsymbol{x}$ constitute a set $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{Ns}\}$, where Ns is the number of Shapelets. $\boldsymbol{X}$ represents the discriminative ability of the Shapelets and supports downstream tasks.

$$\boldsymbol{x} = \begin{cases} \boldsymbol{M} - d^*, & num^0_{NC} \geq num^0_{ASD} \\ d^* - \boldsymbol{M}, & num^0_{NC} < num^0_{ASD} \end{cases} \tag{8}$$

Algorithm 1 shows the pseudocode to facilitate understanding of Brain-Shapelet.

Algorithm 1 Brain-Shapelet

Input: Group-representative brain network $\boldsymbol{F}^{\text{group}}_{i,j}$, random walk length k,BOLD signal set for subjects $\boldsymbol{T}_{Nv} = \{\boldsymbol{t}_{Nv,1}, \ldots, \boldsymbol{t}_{Nv,N}\}$,
Output: $\boldsymbol{X}$: Shapelet distance feature set

```
Repeat:
    Perform random walk to obtain a brain region set W = {w_1, ..., w_k} of length k
    Extract the subsequence set S under W from T_Nv
    Initialize X
    for u = 1 to U do
        Initialize a distance set M
        Initialize a Shapelet candidate S^u_{W,i} from position i in S
        for all r = 1 to U do
            Compute the Euclidean distance m_{u,r} between the subsequence set S^r_{W,n} and the Shapelet candidate S^u_{W,i}.
            Add m_{u,r} to M
        end for
        Compute the optimal splitting point d* and divide M based on Gain(•)
        if the classification accuracy of d* > h do
            if num^0_NC ≥ num^0_ASD do
                x = M − d*
            else do
                x = d* − M
                Add x to X
            end if
        end if
    end for
Return X
```

B. Feature Selection Strategy

Shapelets reflect the instantaneous abnormalities in brain activity and may appear in multiple subjects, resulting in excessively similar redundant features in $\boldsymbol{X}$. It is necessary to remove redundant features and retain only the most important ones before passing $\boldsymbol{X}$ into downstream tasks.

The importance of Shapelets is assessed by their generalizability and their dependency on disease labels. Shapelets extracted from the same brain region set in greater numbers indicate higher generalizability. Jaccard similarity coefficient $J(\cdot)$ is adopted to calculate the element similarity between brain region sets $\boldsymbol{W}_i$ and $\boldsymbol{W}_j$, which provides the generalizability score $\boldsymbol{Ja}_i$ for Shapelet i.

$$J(\boldsymbol{W}_i, \boldsymbol{W}_j) = \frac{|\boldsymbol{W}_i \cap \boldsymbol{W}_j|}{|\boldsymbol{W}_i \cup \boldsymbol{W}_j|} \tag{9}$$

$$\boldsymbol{Ja}_i = \sum_{j=1}^{Ns} J(\boldsymbol{W}_i, \boldsymbol{W}_j) \tag{10}$$

The mutual information $\boldsymbol{I}_i$ measures the dependence between $\boldsymbol{x}$ and the disease labels $\boldsymbol{Y}$.

$$\boldsymbol{I}_i = \sum_{a \subset \boldsymbol{x}_i} \sum_{y \subset \boldsymbol{Y}} p(a, y) \log\left(\frac{p(a, y)}{p(a)p(y)}\right) \tag{11}$$

where $p(a, y)$ represents the joint probability distribution of a and y, while p(a) $p(a)$ and $p(y)$ represent the marginal probability distributions of a and y, respectively.

The generalizability score $\boldsymbol{Ja}_i$ and mutual information $\boldsymbol{I}_i$ are normalized to the same scale to obtain $\boldsymbol{Ja}^{norm}_i$ and $\boldsymbol{I}^{norm}_i$. The normalization process is described as Equation (12)-(13):

$$\boldsymbol{Ja}^{norm}_i = \frac{\boldsymbol{Ja}_i - min(\boldsymbol{Ja})}{max(\boldsymbol{Ja}) - min(\boldsymbol{Ja})} \tag{12}$$

$$\boldsymbol{I}^{norm}_i = \frac{\boldsymbol{I}_i - min(\boldsymbol{I})}{max(\boldsymbol{I}) - min(\boldsymbol{I})} \tag{13}$$

where $max(\cdot)$ and $min(\cdot)$ return the maximum and minimum values respectively from the given data.

The normalized values are combined to compute the importance score $\boldsymbol{imp}_i$ of Shapelet i, which helps to eliminate redundant Shapelets and reduce dimensionality.

$$\boldsymbol{imp}_i = \boldsymbol{Ja}^{norm}_i \times \boldsymbol{I}^{norm}_i \tag{14}$$

C. Brain Region Roles

A series of metrics are defined to measure the roles of brain regions on extracting Shapelet and their associations with other

TABLE I
ABIDE-I DEMOGRAPHIC INFORMATION

SITE	ASD				NC			
	Sex (m/f)	Age Avg(SD)	FIQ Avg(SD)	Handedness (I/m/r)	Sex (m/f)	Age Avg(SD)	FIQ Avg(SD)	Handedness (I/m/r)
NYU	65/10	14.7(7.4)	107.1(16.4)	0/5/14	74/26	15.7(6.2)	113.0(13.4)	0/0/100
UM	57/9	13.2(2.4)	105.4(17.1)	7/1/58	56/18	14.8(3.6)	107.9(9.7)	9/0/65
UCLA	48/6	13.0(2.5)	100.4(13.5)	6/0/48	38/6	13(1.9)	106.4(11.1)	4/0/40
USM	46/0	23.5(8.3)	99.7(16.6)	0/0/46	25/0	21.3(8.4)	115.4(15.1)	0/0/25
PITT	25/4	19.0(7.3)	110.2(14.6)	3/0/26	23/4	18.9(6.6)	110.1(9.4)	1/0/26
YALE	20/8	12.7(3.0)	94.6(21.6)	6/0/22	20/8	12.7(2.8)	105(17.4)	4/0/24
MAX_MUN	21/3	26.1(14.9)	109.1(14.1)	2/0/22	27/1	24.6(8.8)	111.8(9.3)	0/0/28
KKI	16/4	10.0(1.4)	97.9(17.5)	1/3/16	20/8	10.0(1.2)	112.1(9.4)	2/3/23
TRINITY	22/0	16.8(3.2)	108.9(15.5)	0/0/22	25/0	17.1(3.8)	110.9(12.2)	0/0/25
STANFORD	15/4	10.0(1.6)	110.7(16.1)	3/1/15	16/4	10.0(1.6)	112.1(15.4)	0/2/18
CALTECH	15/4	27.4(10.3)	107.7(12.4)	0/5/14	14/4	28(10.9)	114.8(9.6)	1/3/14
SDSU	13/1	14.7(1.8)	111.4(18.0)	1/0/13	16/6	14.2(1.9)	108.1(10.5)	3/0/19
OLIN	16/3	16.5(3.4)	111.3(17.7)	4/0/15	13/2	16.7(3.6)	113.9(16.5)	2/0/13
LEUVEN	26/3	17.8(5.0)	109.4(13.1)	3/0/26	29/5	18.2(5.1)	114.8(12.9)	4/1/29
SBL	15/0	35(10.4)	N/A	1/0/14	15/0	33.7(6.6)	N/A	0/0/15
CMU	11/3	26.4(5.8)	114.5(11.6)	1/1/12	10/3	26.8(5.7)	114.6(9.7)	0/1/12
OHSU	12/0	11.4(2.2)	105.5(21.1)	1/0/11	14/0	10.1(1.1)	115.1(11.1)	0/0/14

FIQ: FULL INTELLIGENCE QUOTIENT , AVG: AVERAGE, SD: STANDARD DEVIATION, M/F: MALE/FEMALE, I/M/R: LEFT/MIXED/RIGHT

brain regions. These metrics also assist in identifying brain regions with potential diagnostic value for ASD.

The occurrence rate $\boldsymbol{Z}_i$ measures the degree of brain region i involvement of each brain region in the Shapelet extraction process. A higher $\boldsymbol{Z}_i$ indicates brain region i is more frequently captured in the random walk process, suggesting it may contain discriminative information.

$$\boldsymbol{Z}_i = \frac{\sum_{q}^{Ns} \delta(i \in \boldsymbol{W}_q)}{Ns} \tag{15}$$

where $\boldsymbol{W}_q$ is the brain region set corresponding to the Shapelet q. The indicator function $\delta(i \in \boldsymbol{W}_q)$ equals 1 if brain region i belongs to $\boldsymbol{W}_q$, and 0 otherwise.

The co-occurrence rate $\boldsymbol{R}_{i,j}$ measures how often both brain regions appear together in the same brain region set, normalized by the total number of occurrences of brain region i in the Shapelet extraction process. A higher $\boldsymbol{R}_{i,j}$ implies a more stable association between brain regions i and j, which frequently co-occur in brain region sets.

$$\boldsymbol{R}_{i,j} = \frac{\sum_{q}^{Ns} \delta(i \in \boldsymbol{W}_q, j \in \boldsymbol{W}_q)}{\sum_{q}^{Ns} \delta(i \in \boldsymbol{W}_q)} \tag{16}$$

The Gini coefficient $\boldsymbol{Gini}_i$ measures the distribution of co-occurrence rates, allowing us to infer the role of each brain region in brain activity. A higher $\boldsymbol{Gini}_i$ value indicates that co-occurrence of a brain region are concentrated among fewer brain regions, suggesting that these associations may contain biological markers. Conversely, a lower $\boldsymbol{Gini}_i$ value implies that the co-occurrence rates are more evenly distributed as Equation (17), indicating that the brain region likely serves as a connector or intermediary in linking multiple brain regions within a brain network.

$$\boldsymbol{Gini}_i = \frac{\sum_{q=1}^{Nv} \sum_{p=1}^{Nv} \left| \boldsymbol{R}_{i,q} - \boldsymbol{R}_{i,p} \right|}{2 \times Nv \sum_{q=1}^{Nv} \boldsymbol{R}_{i,q}} \tag{17}$$

III. EXPERIMENTS

A. Data

The effectiveness of Brain-Shapelet was evaluated using resting-state fMRI data from the Autism Brain Imaging Data Exchange (ABIDE-I) platform [30]. This open multimodal data repository collects data from 17 sites worldwide, including information from 1112 subjects. The fMRI data were preprocessed by the widely adopted Configurable Pipeline for the Analysis of Connectomes (C-PAC) [31], after which the brain was parcellated into 116 regions of interest (ROIs) based on the AAL template [32]. There were 77 subjects excluded after preprocessing, due to missing imaging data or severe motion artifacts, resulting in a final sample of 1,035 subjects (505 in the ASD group and 530 in the NC group). TABLE I provides detailed demographic information for the ABIDE-I dataset and subjects from each site. Further details about the preprocessing process can be found on the ABIDE-I link https://preprocessedconnectomes-project.org/abide/.

B. Comparative Methods

Brain-Shapelet was compared with several state-of-the-art brain network modeling methods to evaluate its effectiveness in ASD diagnosis. These comparable methods include two mainstream strategies: Static Functional Connectivity (SFC)-based models and Dynamics Functional Connectivity (DFC)-based models. In addition, some advanced methods that construct brain networks with time-series or fMRI image data were collectively referred to as "Other methods." The following provides a brief overview of brain network modeling methods.

1) SFC-Based Models: PC-SVM [33] computes the Pearson correlation FC matrix, normalizes with graph Laplacian, extracts feature subspace, and applies SVM for classification.

MVS-GCN [34] constructs multi-view brain networks with thresholds, applies graph structure learning with multi-task embedding for consistent individual representations.

FC-KNN [35] integrates full connectivity and 1-nearest-neighbor FC to constructs multi-graph brain networks with adaptive learning and discriminative constraints.

PLSNet [36] integrates attention mechanisms and positional embedding to construct FC networks and identify key brain regions.

MAHGCN [37] constructs multi-scale FCNs with multi-scale brain atlases, graph convolution, and hierarchical pooling for enhanced classification and interpretability.

2) DFC-Based Models: BrainTGL [38] segments BOLD signals with sliding windows, computes Pearson correlations for each window to construct DFC and capture temporal variations.

MDGL [39] constructs DFCs across multiple brain graphs, aggregates information with GIN and Transformer modules for cross-scale spatiotemporal modeling.

DG-DMSGCN [40] constructs a dual-graph structure with intra-group strong and inter-group differential connections, integrating shared and differential information across DFC segments with dual graph convolution to capture key features under multi-site heterogeneity.

SA-DNN [41] applies Kendall's rank correlation to identify important DFC, and employs multi-head attention for high-order spatiotemporal representation to enhance ASD pattern recognition.

GNN-LSTM [42] combines sliding windows with Ledoit-Wolf shrinkage covariance estimation to construct DFC, enhancing robustness in connectivity estimation.

3) Other Methods: ST-ASDNet [43] works as a dual-path Transformer model for extracting spatial and temporal features to capture ASD-related spatiotemporal dependencies.

ST-Transformer [44] introduces linear multi-head attention to model spatial-temporal dependencies and extract global interaction features.

CNNG [45] processes raw 4D fMRI data with 3D CNNs for spatial feature extraction and GRU for temporal dependencies to capture spatiotemporal evolution.

STCAL [46] encodes fMRI signals with spatial and temporal perspectives, identifies critical regions and time segments via collaborative attention.

BSIGNN [8] learns causal graph structures from time series, constructs directed brain networks with causal modeling and low-rank constraints.

C. Experiments Settings

Model effectiveness of Brain-Shapelet is evaluated via the metrics including classification accuracy (ACC), sensitivity (SEN), specificity (SPE), and the area under the ROC curve (AUC).The parameters of Brain-Shapelet were set as follows: $\beta = 0.3$, $\sigma = 0.6$, $e = 6000$, $k = 7$, $n = 20$. The SVM with a linear kernel function served as the classifier. For the comparative methods, the source codes were from their respective publicly available repositories and implemented with the optimal parameter configurations reported in their original papers.

TABLE II
PERFORMANCE COMPARISON (%) OF BRAIN-SHAPELET AND STATE-OF-THE-ART FCN-BASED METHODS

Category	Method	ACC%	SEN%	SPE%	AUC%
SFC	PC-SVM	63.22	61.25	65.73	63.15
	MVS-GCN	68.93	69.04	63.14	69.14
	FC-KNN	71.65	65.15	57.30	76.09
	PLSNet	72.46	71.61	71.36	75.87
	MAHGCN	72.73	69.83	75.45	72.65
DFC	BrainTGL	67.81	62.67	72.93	67.64
	MDGL	72.50	71.58	66.52	74.12
	DG-DMSGCN	73.13	73.35	72.52	74.51
	SA-DNN	76.23	65.51	83.54	81.35
	GNN-LSTM	80.41	**82.45**	79.34	84.25
Other	ST-ASDNet	67.80	63.32	72.09	70.42
	ST-Transformer	70.77	69.81	71.24	75.21
	CNNG	72.76	67.35	80.21	77.34
	STCAL	73.22	69.58	74.3	73.9
	BSIGNN	74.83	77.51	71.94	78.36
Proposed	Brain-Shapelet	**82.84**	81.66	**84.05**	**89.28**

In the comparative experiments, a 10-fold cross-validation was adopted to assess the performance of Brain-Shapelet. In this setup, the dataset was randomly divided into 10 subsets, with 9 subsets as training data and 1 subset as test data in each iteration, repeating the process 10 times. This method mitigates overfitting during training and ensures a more reliable performance assessment. All methods were trained and tested on the same dataset.

Leave-one-site-out [47] cross-validation was introduced to evaluate the cross-site generalizability of Brain-Shapelet and its applicability to patient populations from different sites. Specifically, in each iteration, 1 site was selected as the test set, while the remaining 16 sites were for training. This strategy effectively simulates real-world deployment scenarios involving unseen sites, thereby assessing the model's generalization performance under varying data acquisition conditions.

D. Experiments Result

1) Model Comparison Experiment: According to the results in TABLE II. Brain-Shapelet achieved competitive performance across all evaluation metrics. Specifically, the framework achieved an ACC of 82.8%, a SEN of 84.0%, a SPE of 81.6%, and an AUC of 89.2%, significantly outperforming other methods. Compared to traditional methods that use Pearson correlation coefficients to construct static brain networks and also use SVM as the classifier, Shapelet improves accuracy by 19.6%. This result strongly demonstrates the effectiveness of Brain-Shapelet on capturing instantaneous abnormalities for ASD diagnosis.

2) Multi-Site Experiment: As shown in TABLE III, although the average performance of Brain-Shapelet slightly declined in the multi-site evaluation, it still maintained competitive accuracy across different sites. The model achieved its highest classification accuracy (ACC) of 83.49% at the CALTECH

TABLE III
CROSS-SITE PERFORMANCE COMPARISON(%) OF BRAIN-SHAPELET

SITE	ACC%	SEN%	SPE%	AUC%
NYU	80.57	76.00	84.00	83.44
UM_1	81.13	75.47	86.79	86.69
UCLA	80.56	78.05	83.87	85.52
USM	77.46	78.26	76.00	87.04
PITT	82.14	89.66	74.07	92.46
YALE	76.00	78.57	72.43	75.74
MAX_MUN	80.87	70.83	89.29	83.33
KKI	77.08	75.00	78.57	84.46
TRINITY	78.72	81.82	76.00	82.36
STANFORD	79.49	78.95	80.00	87.11
CALTECH	83.49	88.74	77.78	92.49
SDSU	77.78	71.43	81.82	83.60
OLIN	82.35	73.68	93.33	91.93
LEUVEN	75.86	64.29	86.67	82.38
SBL	80.00	86.87	73.33	85.78
CMU	81.48	66.67	85.71	88.10
OHSU	80.77	78.57	83.33	89.88
Mean	79.75	77.23	81.35	86.02

site, and its lowest accuracy of 75.86% at the LEUVEN site. These results suggest that Brain-Shapelet effectively captures and aligns key shape patterns across sites, demonstrates resilience to site-specific variability and background noise, and maintains stable performance across heterogeneous subject populations.

E. Parameter Analysis

The sensitivities of five important hyperparameters were examined: number of the Shapelets e, size of the brain region sets k, and length of the sliding window l, as well as the σ and β for constructing the group-representative brain network. Fig.3-a shows that the model achieves the best ACC when $e = 6000$. The value of e should be sufficiently large to represent enough discriminative information. However, if e is too large, it introduces less representative Shapelets and redundant features, resulting in noise.

The length of sliding window l influences the ability of Shapelets to capture dynamic changes in BOLD signals. A shorter sliding window may result in Shapelets that fail to adequately reflect the dynamic nature of BOLD signals, while a longer sliding window may miss short-term changes and introduce noise. As shown in Fig.3-b, setting the sliding window length $l = 20$ resulted in the best ACC, providing a balance between temporal resolution and dynamic representation.

The size of brain region sets k defines the range of brain region interactions covered in the random walk process. Fig. 3c shows that setting $k = 7$ results in the highest ACC. Smaller k values result in overly localized Shapelets that fail to effectively capture the associations between different brain regions. Conversely, larger k values complicate the associations between brain regions, reducing the discriminative ability of the extracted Shapelets.

As shown in TABLE IV, the parameters σ and β influence the sparsity of the $\boldsymbol{F}^{group}$ matrix. When $\boldsymbol{F}^{group}$ is too dense, the random walk tends to generate redundant brain region sets, thereby increasing the model's computational complexity.

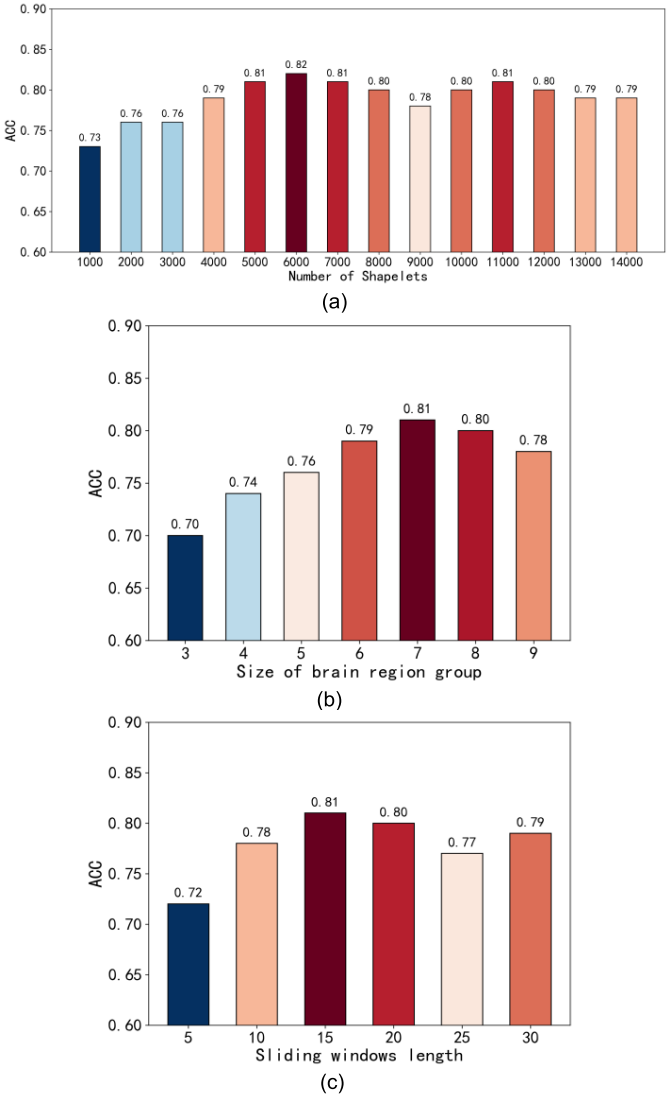


Fig. 3. The impact of key parameters in Brain-Shapelet on classification accuracy.

TABLE IV
NON-ZERO ELEMENT RATIO (%) OF F^{group} ACROSS DIFFERENT SETTINGS OF σ AND β

σ \ β	0.1	0.2	0.3	0.4	0.5
0.1	/	/	/	/	63.0
0.2	/	/	/	67.8	37.8
0.3	/	/	77.8	48.3	23.5
0.4	/	/	62.4	33.4	/
0.5	/	78.4	46.5	22.6	/
0.6	/	62.9	32.1	/	/
0.7	/	44.3	21.0	/	/
0.8	63.4	/	/	/	/

When $\boldsymbol{F}^{group}$ is too sparse, critical discriminative information may be lost. Therefore, σ and β combinations were selected because they resulted in a sparsity range of 20%–80% and conducted five repeated experiments for each setting. In each run, 1000 random walk paths were generated to extract Shapelets. As shown in TABLE V, the parameter setting $\sigma = 0.6$ and

TABLE V
AVERAGE SHAPELET EXTRACTION RATE PER RANDOM WALK (%) (STANDARD DEVIATION)

σ \ β	0.1	0.2	0.3	0.4	0.5
0.1	/	/	/	/	21.2(2.2)
0.2	/	/	/	24.8(2.6)	23.8(4.3)
0.3	/	/	22.5(2.6)	27.1(2.3)	25.5(3.5)
0.4	/	/	23.4(3.4)	25.3(2.8)	/
0.5	/	22.4(1.7)	27.5(4.3)	26.6(1.2)	/
0.6	/	25.2(2.1)	28.3(3.2)	/	/
0.7	/	25.3(2.5)	26.0(3.7)	/	/
0.8	24.7(2.3)	/	/	/	/

$\beta = 0.3$ yielded the highest average Shapelet extraction rate per random walk, reaching 28.3%.

IV. RESULTS ANALYSIS

A. Instantaneous Abnormalities

Fig.4 shows the most frequently observed Shapelet among subjects, which may represent an instantaneous abnormality in brain activity. This Shapelet with a length of 20 was extracted from the following brain regions: CAU.R, PAL.R, THA.L, SFGdor.R, SDGmed.L, OLF.L, and ITG.L. Specific details about brain region divisions and abbreviations can be found from the AAL template [32].

In these regions, fluctuations were weaker and more regular for ASD subjects, but more active and irregular for NC subjects. The Shapelets, appearing frequently among ASD subjects, may represent a type of instantaneous abnormality unique to ASD. It is worth noting that not all ASD subjects displayed this specific instantaneous abnormality. Although there are certain limitations in the applicability of Shapelets, they still provide a level of interpretability that is superior to the methods combining FCN and neural networks.

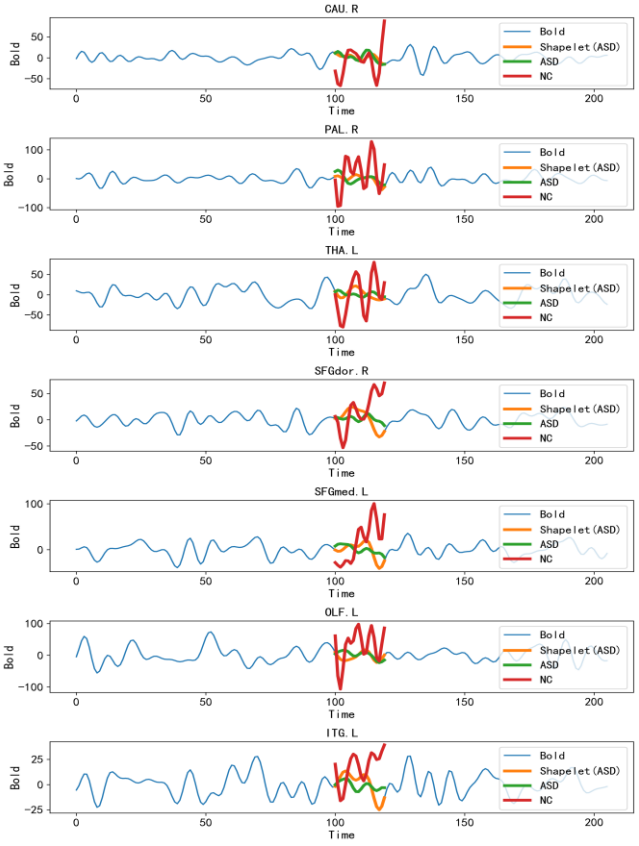


Fig. 4. Instantaneous abnormality represented by the most frequently observed Shapelet. Each subplot shows the best-fit subsequence from the BOLD signals of brain regions corresponding to this Shapelet.

B. Brain Region Roles

The brain region sets corresponding to Shapelets were analyzed to calculate the occurrence rate of each brain region, the co-occurrence rates between associated brain regions, and the Gini coefficients of their distributions.

Fig.5 illustrates the 15 brain regions with the highest occurrence frequencies. They are primarily located in cingulate gyrus (DCG.R, DCG.L, ACG.R), thalamus (THA.L), postcentral gyrus (PoCG.R), basal ganglia (PUT.L, PUT.R), and some brain regions associated with motor and sensory functions (SMA.L, SMA.R). They are involved in cognitive regulation, sensory integration, and motor planning. Functional abnormalities in these regions have been frequently reported in some ASD-related researches [48], [49], [50].

TABLE VI provides detailed co-occurrence rates and Gini coefficients for the 15 most frequently occurring brain regions. There was higher co-occurrence between PUT.L and PAL.R (17.91%), PUT.R and PAL.R (19.53%), and AMYG.L and PUT.L (18.71%). It likely results from their shared characteristics as components of the basal ganglia. Meanwhile, the five brain regions with the highest co-occurrence rates with PUT.L, namely PAL.R, THA.L, AMYG.L, INS.R, and HES.R—are all

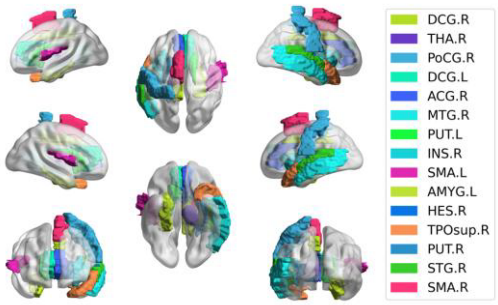


Fig. 5. The 15 brain regions with the highest occurrence frequencies.

associated with emotion processing and transmission. A higher Gini coefficient of PUT.L (0.502) indicates its more stable involvement with these brain regions in instantaneous abnormalities. Such abnormal coordination may have important implications for ASD diagnosis. Previous studies have shown that ASD subjects exhibit hyperconnectivity between PUT.L and INS.R [51], as well as reduced connectivity between the PUT.Land AMYG.L and THA.L [52], providing further support for the abnormal coordination between PUT.L and these frequently co-occurring brain regions in ASD.

HES.R (0.522) is another brain region with a higher Gini coefficient. The associated brain regions—HES.L, PoCG.R, INS.R, PUT.L, and DCG.R—are involved in speech processing, auditory and language functions, and multimodal

TABLE VI
TOP 15 BRAIN REGIONS BY OCCURRENCE RATES, CO-OCCURRENCE RATES, AND GINI COEFFICIENTS

Brain regions	Associated brain regions					Distributions
Occurrence frequencies (%)	Co-occurrence rates (%)	Co-occurrence rates (%)	Co-occurrence rates (%)	Co-occurrence rates (%)	Co-occurrence rates (%)	Gini coefficients
DCG.R (13.84)	DCG.L (14.23)	HES.R (14.05)	ACG.R (13.95)	HES.L (12.64)	PoCG.R (12.41)	0.376
THA.L (13.21)	PUT.L (15.54)	AMYG.L (14.48)	PoCG.R (14.78)	HES.L (14.17)	PAL.R (14.06)	0.434
PoCG.R (12.95)	HES.R (15.59)	THA.L (15.07)	DCG.L (15.04)	DCG.R (13.26)	AMYG.L (12.75)	0.410
DCG.L (12.90)	DCG.R (15.26)	PoCG.R (15.10)	HES.R (14.99)	PUT.L (14.37)	HES.L (14.37)	0.435
ACG.R (12.89)	DCG.R (14.94)	ACG.L (14.78)	PAL.R (14.62)	PUT.L (13.45)	PUT.R (13.45)	0.395
MTG.R (11.96)	PoCG.R (12.63)	THA.L (12.54)	HES.R (12.25)	DCG.R (11.84)	ITG.R (11.75)	0.327
PUT.L (11.81)	PAL.R (17.91)	THA.L (17.38)	AMYG.L (16.76)	INS.R (16.61)	HES.R (16.61)	0.502
INS.R (11.22)	HES.R (17.51)	PUT.L (17.48)	PAL.R (16.58)	DCG.L (15.92)	PUT.R (15.43)	0.473
SMA.L (10.99)	DCG.R (14.54)	PUT.L (14.42)	DCG.L (14.19)	ACG.R (14.00)	PoCG.R (13.88)	0.410
AMYG.L (10.58)	PUT.L (18.71)	THA.L (18.58)	PAL.R (16.69)	DCG.R (16.03)	INS.R (15.39)	0.477
HES.R (10.55)	HES.L (19.46)	PoCG.R (19.13)	INS.R (18.64)	PUT.L (18.60)	DCG.R (18.44)	0.522
TPOsup.R (10.47)	PUT.L (15.30)	INS.R (14.40)	PAL.R (13.76)	DCG.L (13.36)	AMYG.L (13.06)	0.410
PUT.R (10.32)	PAL.R (19.52)	PUT.L (17.94)	ACG.R (16.80)	INS.R (16.77)	THA.L (16.33)	0.495
STG.R (10.16)	PoCG.R (13.83)	PUT.L (13.63)	HES.L (13.59)	PAL.R (13.56)	HES.R (13.14)	0.414
SMA.R (10.11)	DCG.R (13.59)	ACG.R (13.04)	DCG.L (12.48)	PoCG.R (12.27)	HES.L (12.27)	0.372

integration of auditory and somatosensory information. The abnormal coordination of these brain regions may contribute to impairments in language processing and emotional expression in ASD subjects. Previous studies have reported reduced interhemispheric functional connectivity between HES.R and HES.L [53] and decreased resting-state functional connectivity between HES.R and INS.R [54] in ASD subjects. This provides additional support for the existence of abnormal coordination among these brain regions.

V. DISCUSSION

Brain-Shapelet achieves superior performance in ASD diagnosis than FCN-based methods, validating the effectiveness of instantaneous abnormalities. Instead of the global perspective of FCNs, random walk algorithm obtains local brain region sets and independently analyze local information, which is more effective on capturing localized abnormal patterns. Dynamic brain networks often generate high-dimensional connectomes, making it difficult to identify helpful instantaneous features for disease diagnosis. Brain-Shapelet addresses this challenge by selecting instantaneous abnormalities relying on their discriminative ability. It emphasizes short-term alterations in brain activity, effectively reducing data dimensionality and minimizing noise interference, thereby enhancing its ability to capture instantaneous abnormalities. Notably, traditional brain network modeling methods are primarily based on functional connectivity, which focuses on linear associations and often neglects nonlinear patterns between brain regions. In contrast, capturing instantaneous abnormalities explores nonlinear associations between brain regions through time-series analysis, thereby improving the accuracy of ASD diagnosis.

ASD symptoms are diverse and complex with significant variability among subjects. The Shapelets extracted by Brain-Shapelet may only exist in some specific groups of subjects, leading to limited discriminative ability for individual Shapelets. Despite this, brain-Shapelet still achieves excellent performance by combining multiple Shapelets. It proves that Shapelets can effectively capture subtle instantaneous differences in various brain regions between ASD and NC subjects. These differences likely represent hidden or instantaneous abnormalities in brain activity. Nevertheless, the combination of multiple Shapelets also introduces semantic complexity, which limits the interpretability and clinical applicability of the model. The proposed co-occurrence rate and Gini coefficient quantify the cooperative relationships among brain regions involved in Shapelets, helping to identify core brain region sets combinations associated with instantaneous abnormalities.

The associations between brain regions are usually represented by linear functions, but such representations may not directly aid in disease diagnosis. Compared to linear associations, co-occurrence rates measures the probability of two brain regions appearing together within a brain region set during instantaneous abnormalities. A higher co-occurrence rates indicate that these two brain regions frequently co-occur in instantaneous abnormalities, highlighting their collaborative behavior as diagnostically relevant for ASD.

The Gini coefficient measures the distribution of co-occurrence rates. A higher Gini coefficient indicates that the associations between brain regions are more concentrated and stable, implying that the information contained in the BOLD signal of that brain region is more likely to be a key part of the instantaneous abnormalities If a brain region exhibits widespread co-occurrence, it indicates that the region functions as a hub connecting different important brain regions within the brain region set, and its BOLD signal may not be the core part of the instantaneous abnormalities.

This study serves as an initial exploration on Shapelet extraction, classification, and interpretability analysis, while recognizing substantial opportunities for improvement. Our future research could focus on integrating deep learning models to leverage their ability to capture nonlinear patterns, enabling a deeper investigation of high-level feature representations for instantaneous abnormalities. Moreover, the topological relationships between Shapelets could be quantified by graph theory or graph neural networks. Such advancements have the potential to further improve the accuracy and robustness of neurological disorders diagnosis.

VI. Conclusion

We proposed a framework, Brain-Shapelet, for capturing instantaneous abnormalities in brain activity for ASD diagnosis. Traditional FCN-based methods emphasizes global patterns, while Brain-Shapelet focuses on discriminative subsequences of BOLD signals. This framework reduces noise in fMRI data and effectively enhances the ability to capture instantaneous abnormalities. On the ABIDE-I dataset, Brain-Shapelet comprehensively outperformed FCN-based methods. The experimental results demonstrated that Shapelets can effectively capture subtle instantaneous differences in various brain regions between ASD and NC subjects, reflecting hidden or instantaneous abnormalities in brain activity. In the interpretability analysis, the defined evaluation metrics—brain region occurrence rate, co-occurrence rate, and Gini coefficient—revealed that cingulate gyrus, basal ganglia, supplementary motor area, and thalamus repeatedly appeared in the Shapelet extraction process. It indicates that these brain regions play important roles in the extraction of instantaneous abnormalities and this is consistent with prior studies that have implicated these brain regions in ASD. These findings offer a new perspective for identifying potential biomarkers for ASD diagnosis.

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